SMU Lyle School of Engineering



Reinforcement Learning

Dynamic Programming

Sutton/Barto* Chapter 4

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018



Online Material

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Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
  - **Dynamic programming**
  - Monte Carlo methods
  - Temporal-difference learning
  - Multi-step bootstrapping
  - Planning and learning with tabular methods
- Part II: Approximate Solution Methods
  - Prediction and Control using Approximation
  - Eligibility Traces
  - Policy Gradient Methods
- Part III: Modern RL Methods
  - Deep Reinforcement Learning
  - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\text{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s, a)$	expected immediate reward from state s after action a.
$r(s, a, s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Goal

- **Remember**: Finite MDPs have an optimal deterministic policy (under some mild conditions).
- **Goal**: Compute the **optimal deterministic policy** $\pi_* = \underset{a}{\operatorname{argmax}}\, q_*(s,a)$ given a perfect MDP model.
- Solve the Bellman optimality equations (the optimal policy always picks the best action):

$$v_*(s) = \max_a \mathbb{E}_\pi \left[R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a\right]$$

or

$$q_*(s,a) = \max_a \mathbb{E}_\pi \left[R_{t+1} + \gamma \underbrace{\max_{a'} q_*(S_{t+1}, a')}_{= v_*(S_{t+1})} \middle| S_t = s, A_t = a\right]$$

- **Issues**: The value estimate for S_t depends on the current estimates of the successor states S_{t+1}! We need to solve all optimality equations at the same time.

Control Problem:
Find the optimal policy

- **Idea**: We can find the optimal policy by stepwise **improving a policy** till it is optimal. A policy is better if it has larger state values.

$$\pi' \geq \pi \Leftrightarrow v_{\pi'}(s) \geq v_\pi(s) \qquad \text{for all } s \in \mathcal{S}$$

- To do this we need to be able to compute v_π and $v_{\pi'}$.

Prediction Problem:
Estimate the value function for a policy.

Tabular Methods

- Tabular means: The value functions v or q are stored in arrays (tables) called V or Q.
- Important property: Entries in the table do not influence each other. We can change them independently.
- Tabular methods find the exact solution (i.e., optimal value function and policy) given enough memory and time.

- **Issues**:
  - Table size:
    - v and deterministic π have $|\mathcal{S}|$ entries
    - q has $|\mathcal{S}| \times |\mathcal{A}|$ entries

    This is only feasible for problems with a small discrete state/action space!
  - No generalization: Knowledge about the value of one state (state-action pair) does not help us with similar states

- We will later talk about approximate methods that generalize and can work with large state spaces.

Value Function

s	State Value $V(s)$
(1,1)	0.7453
(1,2)	0.8016
…	…

Q-Value Function

s	a	$q(s,a)$
(1,1)	up	0.7453
(1,1)	right	0.6709
(1,1)	down	0.7003
(1,1)	left	0.7109
…	…	…

Policy

s	Action $\pi(s)$
(1,1)	up
…	…
…	…

Prediction vs. Control

- In this class, we will always talk about two tasks:

1. **Prediction to evaluate a policy**: Estimate the value function for a given policy. We can use this to compare two policies.

2. **Control**: Control the agent's behavior to find the optimal policy. This requires being able to evaluate policies.

Prediction: Policy Evaluation

What is the value function for a given policy?

Policy Evaluation to Estimate v_π

Goal: Compute the state-value function v_π for a given policy π.

Bellman expectation equations give us the recursive relationship:

$$\begin{aligned} v_\pi(s) &\stackrel{\text{def}}{=} \mathbb{E}_\pi[G_t|S_t = s] \\ &= \mathbb{E}_\pi\left[R_{t+1} + \gamma G_{t+1}|S_t = s\right] \\ &= \mathbb{E}_\pi\left[R_{t+1} + \gamma v_\pi(S_{t+1})|S_t = s\right] \\ &= \sum_a \pi(a|s) \sum_{s',r} p(s', r\,|s, a)[r + \gamma v_\pi(s')] \end{aligned}$$

with $|\mathcal{S}|$ equations, one for each state s.

Guarantees:

- Discounted finite MDPs ($\gamma < 1$) have a unique v_π.
- Undiscounted finite MDPs ($\gamma = 1$) have a unique v_π if π guarantees to reach a terminal state.
- Finite-horizon MDPs have unique v_π^k where k is the time step. I.e. , value depends on how close we are to the end of the episode.

Issue: Recursive relationship between $v_\pi(s)$ and $v_\pi(s')$

Solution Option 1: Linear Program

Solve the system of $|\mathcal{S}|$ linear Bellman Expectation Equations:

$$v_\pi(s_1) = \sum_a \pi(a|s_1) \sum_{s',r} p(s', r\,|s_1, a)[r + \gamma v_\pi(s')]$$

$$v_\pi(s_2) = \sum_a \pi(a|s_2) \sum_{s',r} p(s', r\,|s_2, a)[r + \gamma v_\pi(s')]$$

… for all $s \in \mathcal{S}$

Formulate as a linear program

- Objective function: $\min_V \sum_{s\in\mathcal{S}} V(s)$
- Constraints: $V(s) \geq \sum_{s',r} p\big(s', r|s, \pi(s)\big)[r + \gamma V(s')] \quad \forall s \in \mathcal{S}$

Why this works: The smallest feasible solution will satisfy all Bellman equations.

$$V(s) = \sum_{s',r} p\big(s', r|s, \pi(s)\big)[r + \gamma V(s')] \quad \forall s \in \mathcal{S}$$

Example: 2-State MDP

Min: $V(s_1) + V(s_2)$

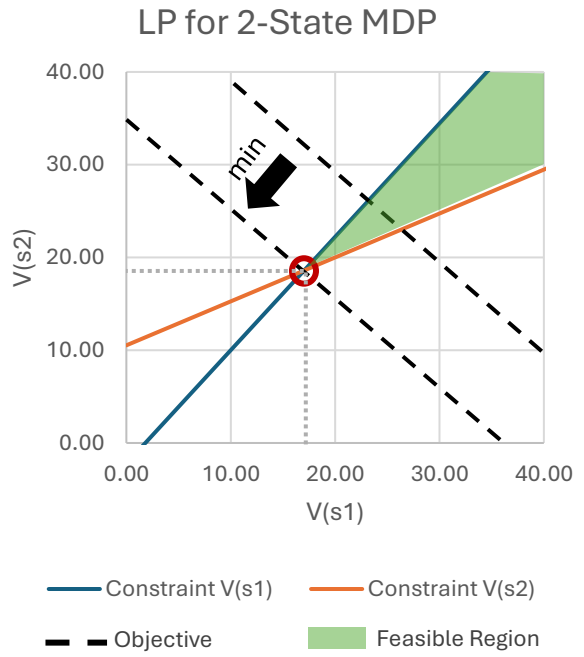
S.t.: $V(s_1) \geq R_1 + \gamma\big(P_{11}V(s_1) + P_{12}V(s_2)\big)$

$V(s_2) \geq R_2 + \gamma\big(P_{21}V(s_1) + P_{22}V(s_2)\big)$

Where **following π** has

R_x immediate reward of s_x

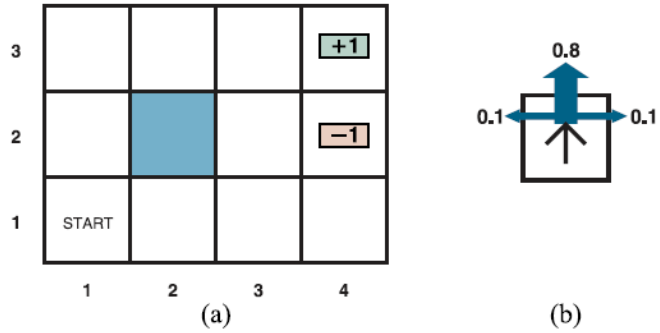
P_{xy} transition probability $s_x \rightarrow s_y$



Solution Option 2: Iterative Policy Evaluation

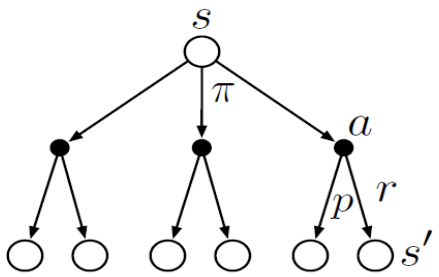
Dynamic Programming: solves a complex optimization problem by breaking it down into smaller, overlapping subproblems and using their solutions to build the overall optimal solution.

Sweep the state space



- Start with arbitrary values for all $v(s)$
- Update all state values (called sweeping over the state space) using the **Bellman Update** (k is the iteration counter):

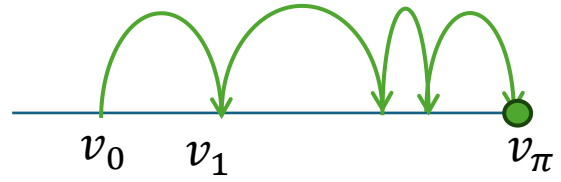
Backup diagram



$$v_{k+1}(s) \stackrel{\text{def}}{=} [R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s]$$
$$= \sum_a \pi(a|s) \sum_{s',r} p(s', r\,|s, a)[r + \gamma v_k(s')]$$

In DP called “expected update.”
Uses the expectation over all possible actions and next states.

Bellman updates



Distance means difference

- **Convergence**: Updates converge to v_π for $k \to \infty$ iterations. Reason is on the next slide.

Convergence of Iterative Policy Evaluation

- The Bellman update can be rewritten as an operator:
  Bellman Expectation Operator T^π:

$$(T^\pi v)(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r\,|s, a)[r + \gamma v(s')]$$

Notation:
$(T^\pi v)$ is the new value function after applying the operator. We could write $v_{k+1} = T^\pi(v_k)$

- **The Fixed Point Property:** The value function is unchanged if all state values are already consistent.

$$v_\pi = T^\pi(v_\pi)$$

- **The Contraction Property**: The operator T^π is γ-contracting (see textbook for derivation):

$$\|T^\pi u - T^\pi v\|_\infty \leq \gamma \|u - v\|_\infty \quad \text{for any} \quad \gamma < 1$$

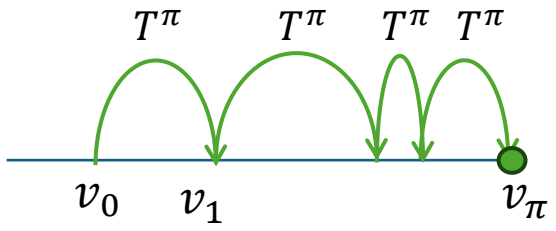
  I.e. : Applying T^π reduces the largest difference (infinity norm) between any arbitrary value functions u and v by at least a factor of γ

- This implies:

$$\|v_\pi - T^\pi v\|_\infty \leq \gamma \|v_\pi - v\|_\infty \quad \text{for any} \quad \gamma < 1$$

Convergence if $\gamma < 1$:
$T^\pi T^\pi T^\pi \ldots T^\pi v \rightarrow v_\pi$ for any v

Bellman updates



Distance means difference

Alg. Iterative Policy Evaluation

Notation: V is the tabular estimate of v

Iterative Policy Evaluation, for estimating $V \approx v_\pi$

Input π, the policy to be evaluated
Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation
Initialize $V(s)$ arbitrarily, for $s \in \mathcal{S}$, and $V(terminal)$ to 0

Loop:
  $\Delta \leftarrow 0$
  Loop for each $s \in \mathcal{S}$:
    $v \leftarrow V(s)$
    $V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s', r \mid s, a)\left[r + \gamma V(s')\right]$
    $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$

Bellman operator T^π: $V_{k+1} = T^\pi V_k$

Stop when value function changes little:
$\|T^\pi V_k - V_k\|_\infty < \theta$
It can be shown that the error is bounded:
$\|V_k - v_\pi\|_\infty < \frac{\theta}{1-\gamma}$

If we stop policy evaluation early (after n iterations) then it is called **modified policy iteration** and we end up with an approximation.

Control: Policy Iteration

Find the optimal policy based on the environment model.

Policy Improvement and the Greedy Policy

Policy Improvement Theorem

$$q_\pi\big(s, \pi'(s)\big) \geq v_\pi(s) \;\; \forall s \in \mathcal{S} \;\; \Rightarrow \;\; v_{\pi'}(s) \geq v_\pi(s) \;\; \forall s \in \mathcal{S}$$

If π' is the same as π but has a better action for at least one state, then π' is strictly better than π

- **Greedy policy improvement**: create a new policy π' by choosing in each state the best (aka greedy) action with respect to $q_\pi(s, a)$.

$$\pi'(s) = \underset{a}{\operatorname{argmax}}\, q_\pi(s, a)$$

- **Guarantee**: This **greedy policy** π' is strictly better unless π is already optimal.
- **Proof**:

$$\begin{aligned} v_{\pi_{k+1}}(s) = q_{\pi_k}\big(s, \pi_{k+1}(s)\big) &= q_{\pi_k}(s, \operatorname{argmax}_a q_{\pi_k}(s, a)) \\ &= \max_a q_{\pi_k}(s, a) \\ &\geq q_{\pi_k}\big(s, \pi_k(s)\big) \\ &\geq v_{\pi_k}(s) \end{aligned}$$

π_{k+1} chooses greedy action for s

Choosing the best action will never lead to a lower value!

Policy Iteration

Idea: Create a sequence of monotonically improving policies:

$$\pi_0 \overset{E}{\rightarrow} v_{\pi_o} \overset{I}{\rightarrow} \pi_1 \overset{E}{\rightarrow} v_{\pi_1} \overset{I}{\rightarrow} \pi_2 \overset{E}{\rightarrow} v_{\pi_2} \overset{I}{\rightarrow} \pi_3 \overset{E}{\rightarrow} v_{\pi_3} \overset{I}{\rightarrow} \dots \pi_* \overset{E}{\rightarrow} v_*$$

$\overset{E}{\rightarrow}$ policy eval. step
$\overset{I}{\rightarrow}$ policy improvement step

Guarantees:

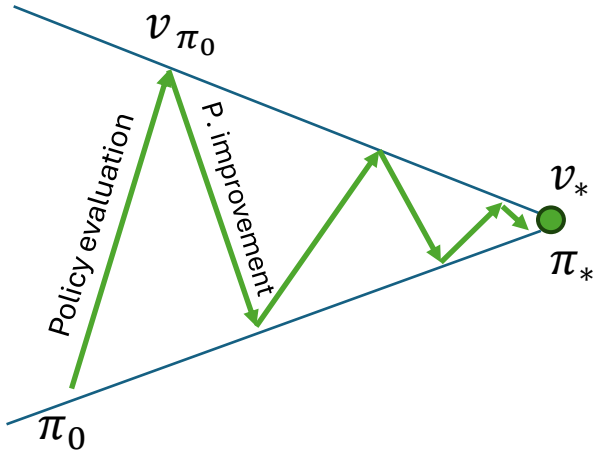
- Policy improvement is guaranteed to improve the policy.
- Finite MDPs have a finite number of policies → Policy Iteration must **converge**!

Advantages:

- Policy becomes stable before v_* is reached.
- Policy iteration typically converges fast.

Issue:

- Policy estimation is computationally expensive!
  Using fewer iterations leads to modified policy iteration.



Alg. Policy Iteration

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$; $V(terminal) \doteq 0$

2. Policy Evaluation
   Loop:
      $\Delta \leftarrow 0$
      Loop for each $s \in \mathcal{S}$:
         $v \leftarrow V(s)$
         $V(s) \leftarrow \sum_{s',r} p(s', r \mid s, \pi(s))\big[r + \gamma V(s')\big]$
         $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
      old-action $\leftarrow \pi(s)$
      $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s', r \mid s, a)\big[r + \gamma V(s')\big]$
      If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

Just policy evaluation from before. We can also stop this loop early (modified policy iteration).

Picks the greedy action with the highest Q-value to improve the policy.

Control: Value Iteration

Find the optimal value function based on the environment model.

Value Iteration

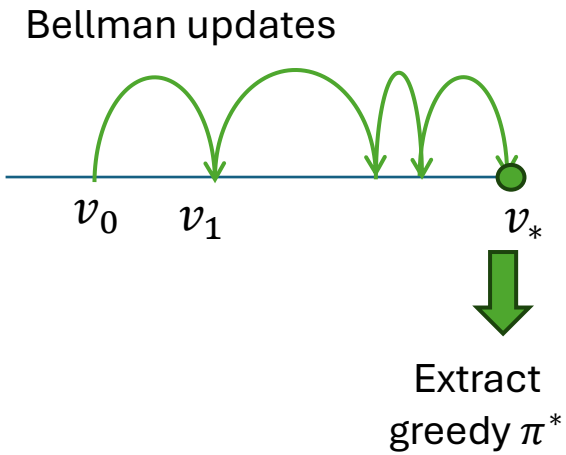
- Combines policy improvement and truncated policy evaluation into a single step. This is called the **expected Bellman optimality update**.

$$v_{k+1}(s) \stackrel{\text{def}}{=} \max_a \mathbb{E}[R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s, A_t = a]$$
$$= \max_a \sum_{s',r} p(s', r\,|s, a)[r + \gamma v_k(s')]$$

$\max_a$ ... policy improvement
$\mathbb{E}$ ... truncated policy evaluation (1 step)

- Written as the **Bellman optimality operator**:

$$(T^* v)(s) = \max_a \sum_{s',r} p(s', r\,|s, a)[r + \gamma v(s')]$$



- **Guarantee**: T^* is a non-linear operator, but it is also γ-contracting. The Sequence $\{v_k\}$ converges to v_*.

Alg. Value Iteration

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$; $V(terminal) \doteq 0$

2. Policy Evaluation
   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
   $V(s) \leftarrow \sum_{s',r} p(s', r|s, \pi(s))\left[r + \gamma V(\right.$
   $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta <$ [illegible]

3. Policy In[illegible]
   policy-sta
   For each
   old-action $\leftarrow \pi(s)$
   $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s', r|s, a)\left[r + \gamma\right.$
   If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$
   If *policy-stable*, then stop and return $V \approx v_*$

Combine improvement and modified evaluation (1 iteration only) into a single update.

Value Iteration, for estimating $\pi \approx \pi_*$

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(terminal) = 0$

Loop:
  $\Delta \leftarrow 0$
  Loop for each $s \in \mathcal{S}$:
    $v \leftarrow V(s)$
    $V(s) \leftarrow \max_a \sum_{s',r} p(s', r|s, a)\left[r + \gamma V(s')\right]$
    $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$

Bellman optimality operator: $V_{k+1} = T^* V_k$

Stopping criterion from policy evaluation

Output a deterministic policy, $\pi \approx \pi_*$, such that
$\pi(s) = \arg\max_a \sum_{s',r} p(s', r|s, a)\left[r + \gamma V(s')\right]$

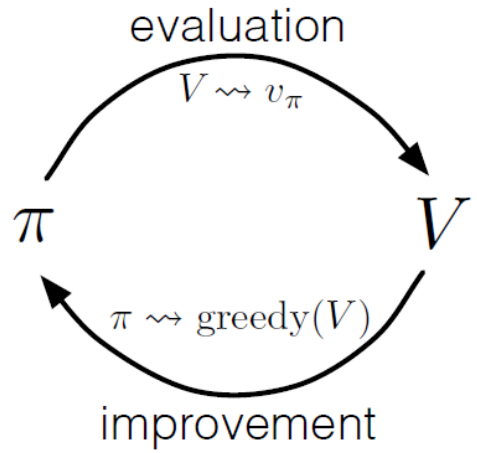
Extract the greedy policy.

Code...

GPI: Generalized Policy Iteration

Generalizing value and policy iteration into a single framework.

GPI: Generalized Policy Iteration

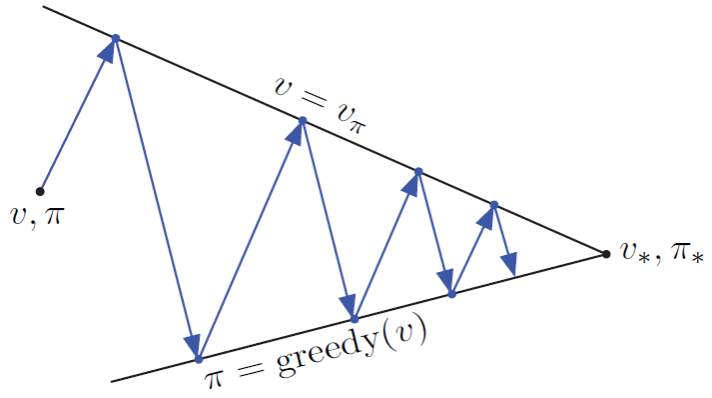


Steps

- **Policy evaluation**: make the value function consistent with the current policy π.
- **Policy improvement**: make policy greedy with respect to the current value function estimate V.

Most DP methods can be described as a special case of GPI:

- **Policy iteration**: each evaluation and improvement step is completed (by a sweep over $\mathcal{S}$) before moving on.
- **Value iteration**: uses a single iteration for the policy evaluation step.
- **Asynchronous DP**: Instead of sweeps over $\mathcal{S}$, updates states in some other order (e.g., focus on states where the value function changed a lot in the last step). This interleaves pol. eval. and pol. Improvement.



All three approaches typically converge to the optimal solution due to the **Policy Improvement Theorem**.

Issues with DP Methods

Require perfect knowledge of the model!

Space Complexity

- Tabular method: stores the complete value function as tables V or Q.

Time Complexity

- Updates repeatedly sweep over the whole state space.
- Wastes computation by updating states that are not relevant for the optimal behavior. I.e., the optimal policy will never visit them.
- Tabular methods can not generalize between similar states.

Asynchronous DP can help

Efficiency

- **Direct Search**: In each state we can choose one of the available actions. Search space $O(|\mathcal{A}|^{|\mathcal{S}|})$ is **exponential** in the number of states $|\mathcal{S}|$.

- **LP (linear program):** Linearize the non-linear Bellman Optimality Equations by introducing constraints for all action/state pairs. Leads to a time complexity of $O(|\mathcal{S}|^3)$.

- **DP**: Time $O(|\mathcal{S}|^2 \times |\mathcal{A}|)$ is **quadratic** in $|\mathcal{S}|$.

Sparse transition matrices lead to better performance.

Conclusion: Search and LP are impractical. The number of actions is typically relatively small. **DP can be used today with millions of states.**

What You Need to Know



Requirements:

- **Model access**: DP methods need access to the **model**'s function p.
- **Small discrete state/action space** since it is a tabular method.

Tasks:

- **Prediction**: Calculates the value function for a given policy called in DP policy evaluation.
- **Control**: Calculates the optimal policy for an MDP. Is not truly control, but an offline planning method.

Value Iteration:

- Convert the recursive Bellman equations into the Belmann update operators and sweep the state space till the optimal value function is found.
- **The Expected Bellman operator is γ-contracting and converges to the optimal value function** (requires either $\gamma < 1$ or a guarantee that the policy will reach a terminal state).

Policy Iteration:

- **Policy Improvement Theorem**: the policy strictly improves with each policy iteration step.
- Value iteration is a special case of modified policy iteration with a single policy evaluation iteration.
- All DP methods can be generalized using the GPI evaluation/improvement framework.